

WristMap

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Brief Overview/Description

Our Project is called “ WristMap”. The WristMap is a travel device that is worn around the wrist. It is convenient and comfortable for daily use, especially for people who travel often. This travel device offers a full screen display of a map. It features GPS navigation which allows you to select specific places such as home or work. It entails real-time traffic updates which allows the GPS system to reroute you to a better route with no traffic. This project will offer people a simple way to explore their surroundings safely and timely.

Requirements

- GPS positioning- A user should be able to get detailed GPS directions and given a map view of the route. A user should be able to use the map view to mark and save any important locations.
- Traffic forecast- Given that a user is on a route, they should be given an updated traffic forecast if it is going to slow down their current route or speed up another. This allows the user to take an alternate path to avoid as much traffic as possible.
- Accessibility- A user should be able to navigate the software through a touch screen interface and with the use of voice commands for ease of use.
- Battery life- A user should be able to use the device for extended periods of time throughout the day without fear of the device running out of battery.
- Testing environment- Testing environments and test cases should be developed so that the application and its functions can be tested.
- Hardware- Required hardware that is needed so that the device can function as intended.
- Expendability and maintenance- The code should be written in a way that allows for the introduction of new features and is easily maintainable for potential bugs introduced.

Implementation Milestones

- a. - Project Approval: Create a proof of concept to present to the client
 - Design implementation: Create designs for each screen of the watch and implement the designs for each screen in the code

- Software implementation: Set up GPS module to support GPS navigation, set up traffic API requests to update traffic forecasts, set up backend database for points of interest storage, and enable voice dictation and commands for navigation.

- Hardware creation: Design and construct watch with touch screen and voice input and research and develop a battery system for long lasting battery life.

- Testing and maintenance: Develop test conditions for software and hardware requirements and then fix any missed bugs or features

- b. Front End and Back End designs for each one: GPS positioning, Traffic forecast, Accessibility, Hardware, Battery Life, Testing environment, Extendability and maintenance

- c. Developer: Implementation planning and requirements, requirement implementation, and bug fixes

- Tester: Test planning, Testing requirements, bug write ups, and bug fixes retesting.

- d. Documentation: Creating Document and writing all the requirements needed. Time spent was a week.

- PowerPoint: Creating and designing PowerPoint that meets all requirements and then recording the slides. Time spent was a week.

Need analysis

Apple Watch with Apple Maps

Description: Apple Watch also contains a built-in navigation system, just like WristMap. It provides all the same features as WristMap such as turn by turn directions, location monitoring, points of interest, etc.

Gaps:

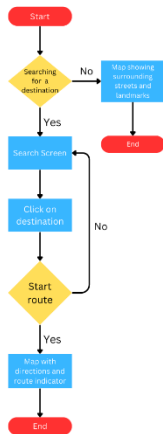
- a. Small screen sizes make usability a little harder and makes it less user friendly.
- b. Requires iPhone pairing to be able to fully function.

Overcoming Gaps:

- a. The WristMap will offer a bigger screen size to offer a more user-friendly device.
- b. The WristMap will be able to fully function without relying on any smartphone.

Architecture

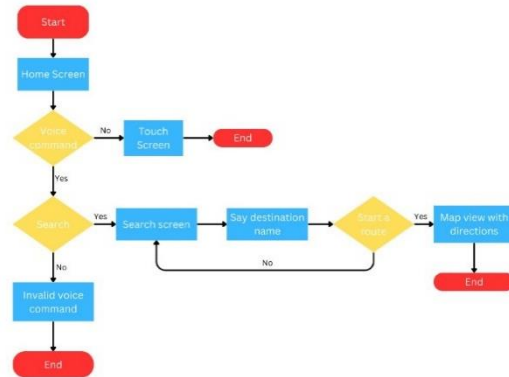
GPS positioning



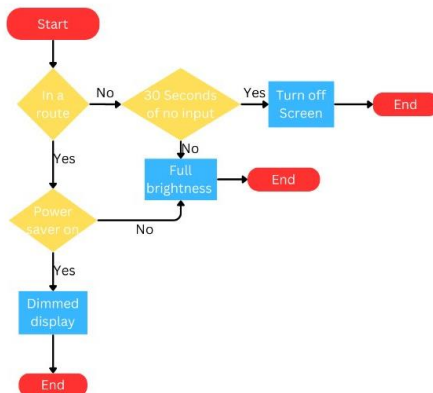
Traffic forecast



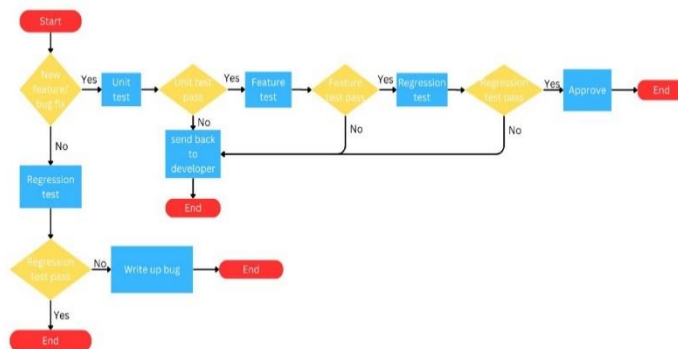
Accessibility



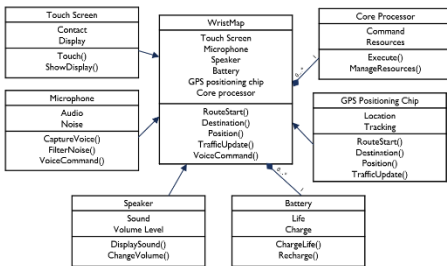
Battery Life



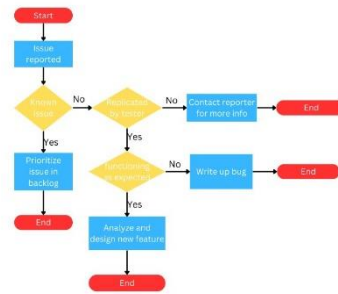
Testing Environment



Hardware

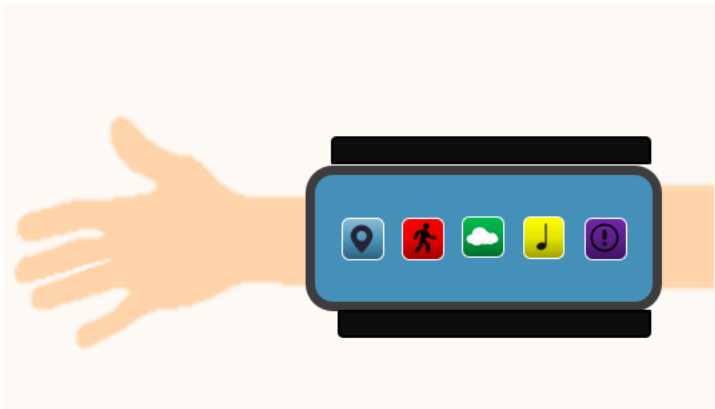


Extensibility and Maintenance



Software Design

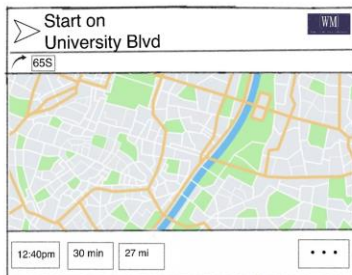
This is the design for the hardware for the WristMap



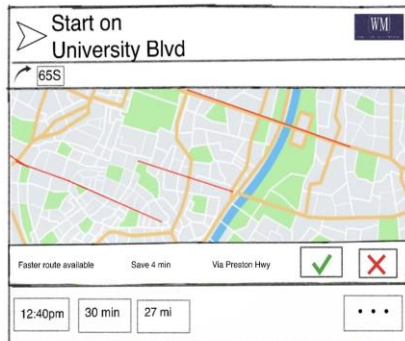
This is the design for the loading screen when you first open the WristMap



This is the design for when you pick your location. It provides the directions at the top and at the bottom it provides an ETA, time it will take, and the miles to get there.



This is the design for when it offers a faster route. You can hit the check or x for yes or no. It also provides how many minutes it will save you and the road you will take instead.



Risk Analysis

- If a code seems to be unclear for the system, we will work together to revision it to the best as possible.
- A piece of hardware could be damaged, so we can manage up a new one soon.
- If the test is not clear, then we will send it back to do some repairs for what's causing trouble.
- If the system seems to be going off course from what it's suppose to do, we'll examine and fix the problem.
- Any bugs in the system will be reported and fixed immediately.

Software Implementation

For our "WristMap" project we plan to use the programming language python. We will need five people for the development. Three for the front-end and two for the backend. The resources needed would be hardware such as computers and laptops that can help run the software needed for it. Software is another resource we will need such as google maps for the GPS, access to the cloud for real time traffic updates, Git, and an IDE.

GPS positioning: 10 days

Traffic forecast: 10 days

Accessibility: 10 days

Hardware: 10 days

Battery life: 5 days

Testing environment: 5 days

Unit testing: 10 days

2 days for each requirement

Extendibility and maintenance: 5 days

Defect Plan: Defects will be located and found. Developers will go through and fix the defects. They will be retested to ensure they are resolved.

Responding to a bug: Developers will receive a bug ticket from the testers, Then they will fix the bug and retest to ensure it is fixed.

Test Plan and Test Script

Functional Testing : We will test each hardware in the system and conduct tests on the functional requirements included in the device to see if they meet the requirements for the customer.

Performance Testing : We'll test the stability, speed, and the conditions of the device to see if it reacts and responds correctly, where it can load quickly and be resilient to much stress.

Regression Testing : In case there is new python code implemented in the device for recent updates and new hardware, we will conduct some testing to the device to see if the new codes causes defects or bugs, and to quickly resolve them.

Alpha and Beta Testing : Once the software is completed and sent out, we will have internal teams, external teams, and developers to run the final diagnostics on the device before the final release.

Testing Data: The testing data that is needed to test our requirements would be destinations and traffic scenarios to test different routing scenarios and the effectiveness of our traffic avoidance system. Voice commands are also needed to test the accessibility of the app.

Reporting a Bug: When a tester comes across a non-critical bug, they should create a ticket with a brief explanation of what the bug is, steps to recreate the bug, and what the expected behavior of that feature should be when the bug is fixed.

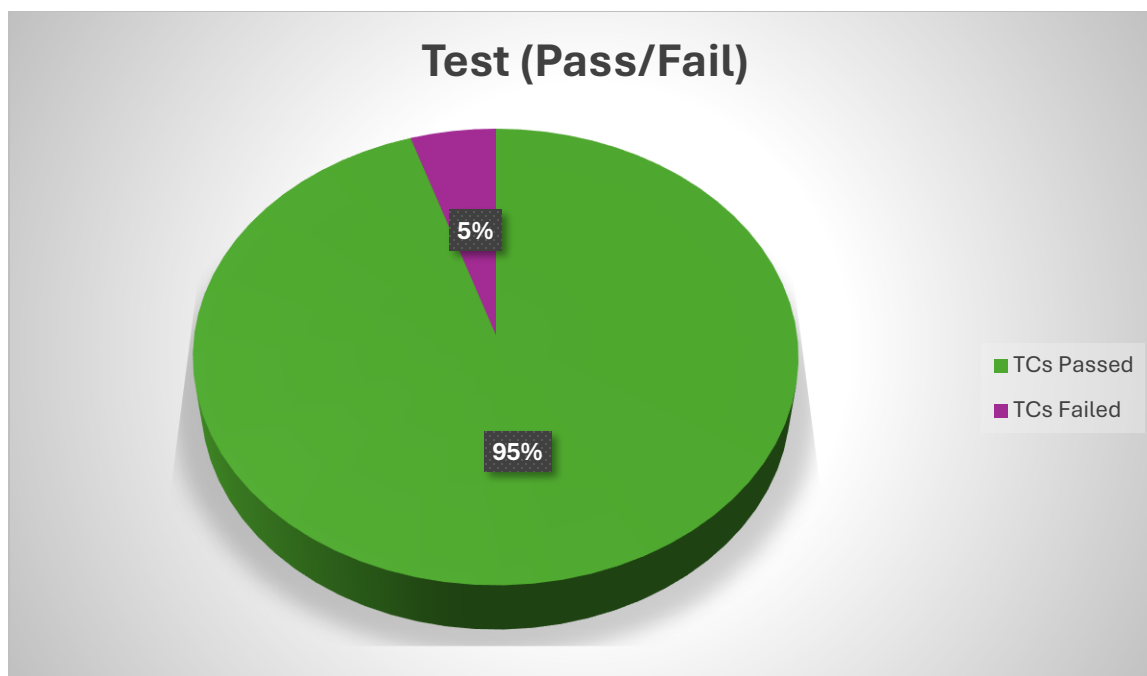
Test Case ID:	TC_01
Test Scenario	FR 1: GPS Positioning
Test Status – Pass/Fail	N/A
Test Data	Destinations
Device	WristMap
Test Step	Expected Results for Each Test Step
A user should be able to see their location on the map	WristMap should display a marker for the location of the user that updates as the move

A user should be able to select a destination	WristMap should move the map to display the selected destination
A user should be able to route to a destination	WristMap should show a route line that outlines the path from the user to the destination

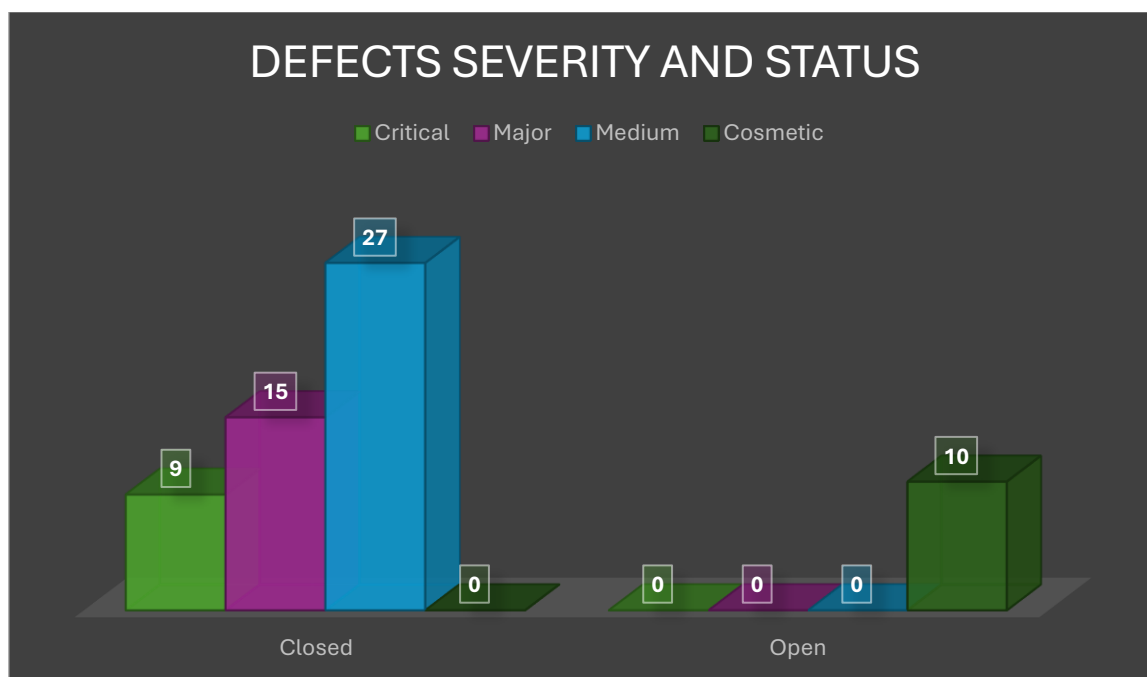
Test Case ID:	TC_02
Test Scenario	FR 3: Accessibility
Test Status – Pass/Fail	N/A
Test Data	Voice Commands: Search, Start Route
Device	WristMap
Test Step	Expected Results for Each Test Step
A user should be able to manipulate the map using their finger	WristMap should move, zoom in, and zoom out of the map
A user should be able to tap on a destination	WristMap should bring up the destination's information when the destination is tapped on
A user should be able to search handsfree	WristMap should bring up the search screen when the voice command: Search is used
A user should be able to start a route handsfree	WristMap should start a route to the selected destination when the voice command: Start Route is used

Test Report

Test Cases Planned	Test Cases Executed	Test Cases Passed	Test Cases Failed
120	84	80	4



	Critical	Major	Medium	Cosmetic	Total
Closed	9	15	27	0	51
Open	0	0	0	10	10
Total					61



Team Assignments and Responsibilities

Kylie Dennison

- Homework 1: Oversee the product's concept and design, create a description and expected requirements of the product for the project approval, create and implement the designs for the software and hardware.
- Homework 2: Designed the front-end examples for FR 1, FR 2, and FR 3, started up PowerPoint and put together Project Overview and Needs Analysis slides.
- Homework 3: Developer, determined a developer plan, determined requirements for implementation, determined time frames for implementation of each requirement, determined a plan for addressing defects and bugs, Put together Project Overview and Software Implementation slides.
- Final Presentation and Document: Started up powerpoint and document. Completed the entire document except for one subheading. Put together Brief Overview, Requirements, Initial Implementation Milestones, and Needs Analysis slides.

Carli Crump

- Homework 1: Create a hardware prototype and Oversee testing and maintenance, create a hardware prototype from provided designs, research and implement the proper battery system, develop test cases for each system requirement and report bugs found.
- Homework 2: Designed back-end charts for FR 1, FR 2, FR 3, FR 4, NFR 1, and NFR 3, put together Implementation Milestones and Team Assignments and Responsibilities slides.
- Homework 3: Tester, determined what testing data was needed to test the requirements, Determined the plan for reporting bugs, determined Implementation milestone timeline for testing, wrote test scripts for two functional requirements, put together Test Scripts, implementation Milestones, and Team Assignments and Responsibilities slides.
- Final Presentation and Document: Put together Software Design, Architecture, Test Plan, Test Script, and Team Assignments and Responsibilities slides. Helped finish document.

Danny El Rayes

- Homework 1: Implement the software requirement, implement required modules and code for frontend development, implement required databases for backend development, fix any missed bugs or features found after testing.
- Homework 2: Designed front end and back end for NFR 2, Worked on Software and Architecture slides, and put together Risk Analysis slide.
- Homework 3: Tester, determined how functional, performance, regression, alpha, and beta testing were going to be conducted, determined number of test cases that were to be ran, determined which functional requirements to write test scripts for, wrote the test report and created visuals for the test cases, put together Test Plan. and Test Report slides.
- Final Presentation and Document: Put together Risk Analysis, Software Implementation, and Test Report slides.

Final Presentation Video

[FinalPresentation.mp4](#)